

Exploring the AKS cluster – Azure infrastructure side



- **Load Balancer (LB)**

- Operates at transport the layer, which is layer 4 of the OSI (Open Systems Interconnection) model, and supports both, inbound and outbound scenarios
- Azure provides two options, public and internal Load Balancer, and two tiers, Basic and Standard
- Used to distribute incoming traffic flows to the backend pool instances from the frontend load balancer
- Two functions of an AKS public load balancer are as follows:
 1. By translating the private IP address to a public IP address that is a part of its Outbound Pool, it provides outbound connections to the cluster nodes inside the AKS virtual network. In other words, it can be used to allow AKS nodes to go to the Internet
 2. It helps expose applications via Kubernetes services types LoadBalancer
- By default, AKS creates a public Load Balancer named “kubernetes” and a public IP address for the first functionality (providing outbound connectivity for nodes)
- If you create a Kubernetes service type LoadBalancer, AKS will automatically configure the Azure Load Balancer called “kubernetes” and will provision a public IP

- **Public IP (PIP)**

- A unique IP address given to a resource in Azure that is reachable from the Internet
- The use case of a PIP in AKS is strongly related to the kubernetes LB and helps to achieve the functionalities mentioned above

Exploring the AKS cluster - Azure infrastructure side

